CS 165B HW 1 4/21

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1. I. a. $E[X] = 1 \cdot 0.6 + 0 \cdot 0.4 = 0.6$

$Var(X) = pq = 0.4 \cdot 0.6 = 0.24$

$E[Y] = E[3X+5] = 3E[X]+5 = 3 \cdot 0.6 + 5 = 1.8 + 5 = \boxed{6.8}$

$Var(Y) = Var(3X+5) = 9Var(X) = 9 \cdot 0.24 = \boxed{2.16}$

b. $X \sim N(0,1)$ $E[X] = 0$ $Var(X) = 1$

$E[Y] = E[2X+3] = 2E[X]+3 = \boxed{3}$

$Var(Y) = Var(2X+3) = 4Var(X) = \boxed{4}$

II. a. $P(spam) = 0.2$ $P(offer|spam) = 0.4$ $P(offer|spam^c) = 0.05$

$P(spam|offer) = \frac{P(offer|spam) \cdot P(spam)}{P(offer)}$

$P(offer) = P(offer|spam) \cdot P(spam) + P(offer|spam^c) \cdot P(spam^c)$

$= 0.4 \cdot 0.2 + 0.05 \cdot 0.8$

$= 0.08 + 0.04 = 0.12$

$P(spam|offer) = \frac{0.4 \cdot 0.2}{0.12} = \boxed{2/3}$

b. $P(A) = \frac{1}{2}$ $P(B) = \frac{1}{2}$ $P(r|A) = 0.4$ $P(b|A) = 0.6$

$P(r|B) = 0.8$ $P(b|B) = 0.2$

$P(B|r) = \frac{P(r|B) \cdot P(B)}{P(r)}$

$P(r) = P(r|B) \cdot P(B) + P(r|A) \cdot P(A)$

$= \frac{1}{2}(0.8 + 0.4) = \frac{1}{2}(1.2) = 0.6$

$P(B|r) = \frac{0.8 \cdot \frac{1}{2}}{0.6} = \frac{0.8}{1.2} = \boxed{2/3}$

III. a. Entropy measures how uncertain or random a distribution is

- high entropy for equally likely outcomes

- $H(P) = -\sum_i p_i \log p_i$

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Cross entropy measures how well distribution Q approximates the true distribution P

- if $Q = P$, cross entropy equals regular entropy
- the worse Q approximates P, the higher the cross entropy

$H(P, Q) = -\sum_i p_i \log q_i$

KL-Divergence measures the "distance" between two distributions

- how much information is lost when you use Q to approximate P

$D_{KL}(P \| Q) = \sum_i p_i \log \frac{p_i}{q_i}$

2. I. a. $\|A\|_p = \sqrt{1^2+2^2+3^2+4^2} = \sqrt{1+4+9+16} = \sqrt{5+25} = \boxed{\sqrt{30}}$

b. $\|B\|_1 = \max(4, 6) = \boxed{6}$

c. $\|C\|_\infty = 5$

II. a. $AB = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 2+2 & 0+6 \\ 6+4 & 0+12 \end{bmatrix} = \boxed{\begin{bmatrix} 4 & 6 \\ 10 & 12 \end{bmatrix}}$

b. No, Matrix multiplication is not commutative

$BA = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2+0 & 4+0 \\ 1+9 & 2+12 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 10 & 14 \end{bmatrix}$

$AB \neq BA$

c. $Ax = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ -1 \end{bmatrix} = \begin{bmatrix} 8-1 \\ 0-3 \end{bmatrix} = \boxed{\begin{bmatrix} 7 \\ -3 \end{bmatrix}}$

III. a. $\frac{\partial f}{\partial x} = 6x + 2y \quad \frac{\partial f}{\partial y} = 2x + 2y$

$\nabla_f = \langle 6x+2y, 2x+2y \rangle$

b. $\|Ax - b\|_2^2 = (Ax-b)^T(Ax-b) = (x^TA^T - b^T)(Ax-b)$

$= x^TA^TAx - x^TA^Tb - b^TAx + b^Tb$

$x^TA^Tb = (x^TA^Tb)^T$ because x^TA^Tb is a scalar

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$(x^T A^T b)^T = b^T A x$

$\Rightarrow \|Ax-b\|_2^2 = x^T A^T A x - 2b^T A x + b^T b$

$\nabla_x (x^T C x) = 2Cx$ when C is symmetric ($A^T A$ is symmetric)

$\nabla_x (b^T A x) = A^T b$

$\nabla_x (b^T b) = 0$

$\nabla_x f(x) = 2A^T A x - 2A^T b = \boxed{2A^T(Ax-b)}$

c. $Wx = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - x_2 \\ 2x_2 \end{bmatrix}$

$W_x\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad \sigma(Wx) = \begin{bmatrix} 0 \\ 2 \end{bmatrix}$

$f(x) = v^T \sigma(Wx)$

$f'(x) = W^T \cdot \mathrm{diag}(\sigma'(Wx)) \cdot v$

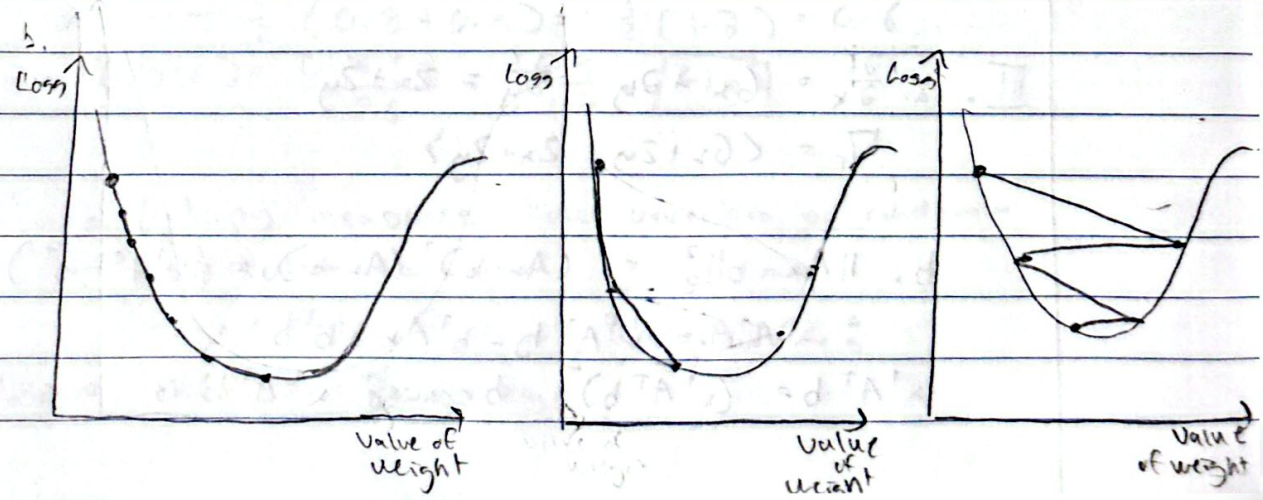
$\sigma'(Wx) = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathrm{diag}(\sigma'(Wx)) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$

$f'\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ -1 & 2 \end{bmatrix} \cdot \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -2 \end{bmatrix}$

$= \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = \boxed{\begin{bmatrix} 0 \\ -4 \end{bmatrix}}$

3. I. a. No, since the term is grad^2, this update rule always decreases w regardless of the sign of the gradient, so if initial weight is to the left of the minimum, it will move farther away from it.

b.



II. a. Model A is likely overfitting because of the large difference in training and validation accuracy, while Model B doesn't have that big of a difference between these two metrics making it better for generalization

b. The specific phenomenon is called overfitting while training when training loss + validation loss start off moving in the same direction and then deviate.

1. Early stopping - stop training at the point where validation loss is at its minimum, before it starts rising again.
2. Reduce model complexity - use a simpler model with fewer parameters so it can't memorize the training data as easily.

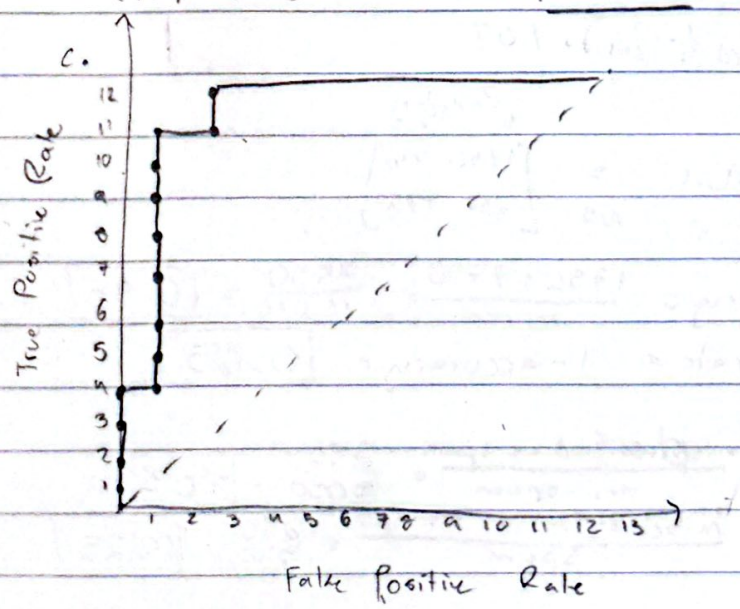
4. I. a. There are 9 ranking errors

- x_{13} appears before 8 positive samples
- x_{15} appears before 1 positive sample

b. $12 \times 13 = 156$ total pairs

error rate = $9/156 \approx \boxed{0.058}$

c.



II. a. $L_{0-1} = \begin{cases} 1 & m < 0 \quad \text{(misclassification)} \\ 0 & m > 0 \quad \text{(correctly classified)} \end{cases}$

$L_{hinge} = \max(0, 1-m) = \begin{cases} 0 & m \geq 1 \quad \text{(confident, correct)} \\ 1-m & m < 1 \quad \text{(unconfident, incorrect)} \end{cases}$

$L_{squared} = \begin{cases} 0 & m > 1 \\ (1-m)^2 & m \leq 1 \end{cases}$

b. 1. $m = \frac{3 \cdot 2 + 2 \cdot 2 + 4 \cdot 3 - 18}{9+4+16} \cdot 1 = \frac{6+4+12-18}{29+16} = \frac{10-6}{29} = \frac{4}{29}$

$L_{0-1}(4/29) = 0$

$L_{hinge}(4/29) = 25/29$

$L_{squared}(4/29) = 0.743$

2. $m = \frac{3 \cdot 3 + 2 \cdot 3 + 4 \cdot 1 - 18}{9+4+16} \cdot -1 = \frac{9+6+4-18}{29} \cdot -1$

$= \frac{15-14}{29} \cdot -1 = -\frac{1}{29}$

$L_{0-1}(-\frac{1}{29}) = 1$

$L_{hinge}(-\frac{1}{29}) = 30/29$

$L_{squared}(-\frac{1}{29}) = 1.07$

III. a. $C = \begin{array}{c|cc} & \text{Predicted S} & \text{Predicted NS} \\ \hline \text{Actual S} & 1750 & 250 \\ \text{Actual NS} & 250 & 7750 \end{array}$

Accuracy $= \frac{1750 + 7750}{10{,}000} = \frac{9500}{10{,}000} = \boxed{0.95}$

error rate $= 1 - \text{accuracy} = \boxed{0.05}$

b. FPR $= \frac{\text{misclassified as spam}}{\text{non-spam}} = \frac{250}{8000} = \boxed{0.03}$

FNR $= \frac{\text{misclassified as non-spam}}{\text{spam}} = \frac{250}{2000} = \boxed{0.125}$

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c. $TPR = \frac{\text{correctly classified spam}}{\text{spam}} = \frac{1750}{2000} = \boxed{0.875}$

$TNR = \frac{\text{correctly classified non-spam}}{\text{non-spam}} = \frac{7750}{8000} = \boxed{0.969}$

d. $\text{Precision} = \frac{1750}{2000} = 0.875$ (correctly predicted spam out of all classified as spam)

$F_1 = 2 \cdot \frac{\text{Recall} \cdot \text{Precision}}{\text{Recall} + \text{Precision}} = \frac{2 \cdot 0.875 \cdot 0.875}{0.875 + 0.875} = 0.875$

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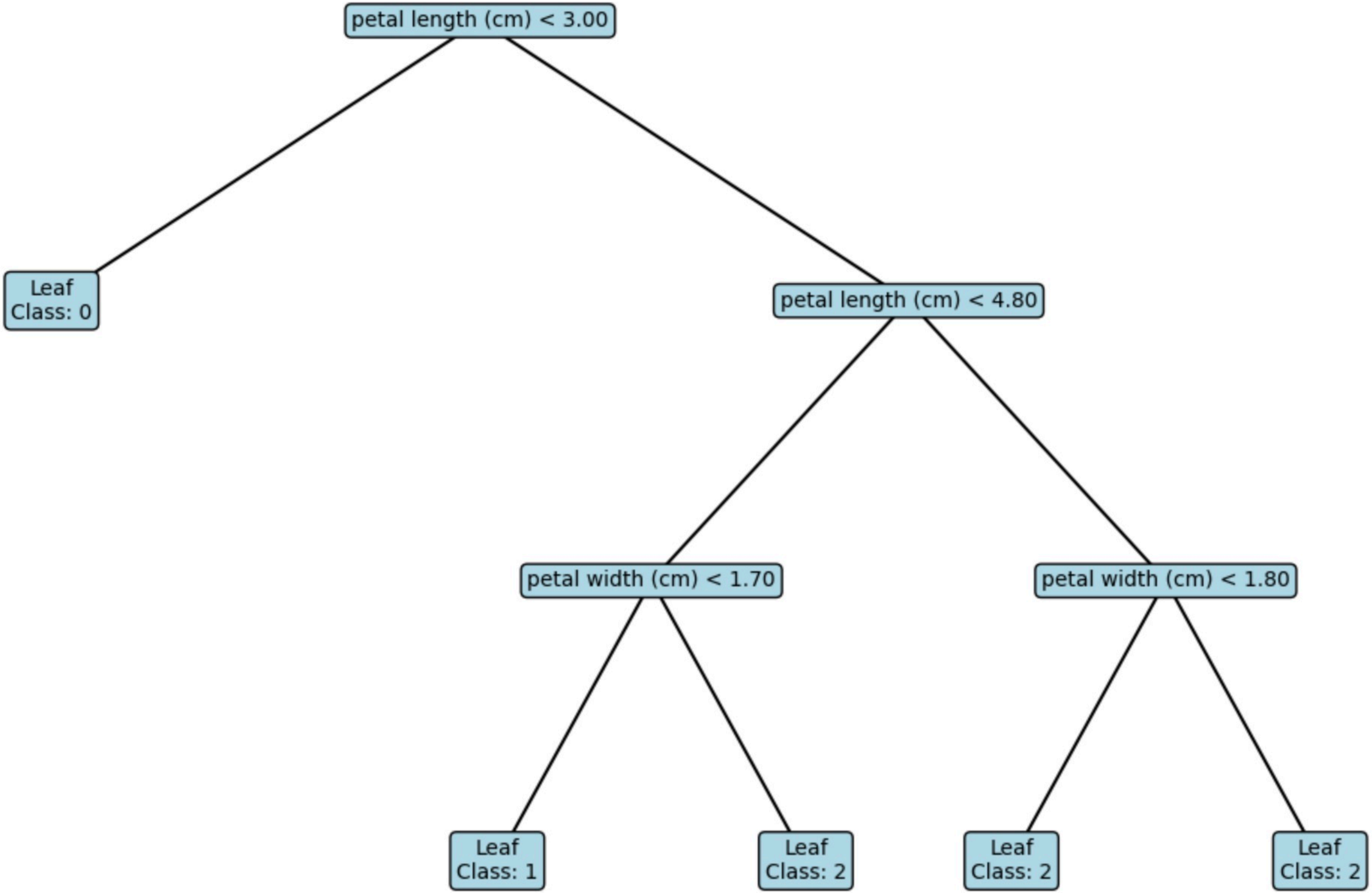
petal length (cm) < 3.00
Leaf
Class: 0
petal length (cm) < 4.80
petal width (cm) < 1.70
petal width (cm) < 1.80
Leaf
Class: 1
Leaf
Class: 2
Leaf
Class: 2
Leaf
Class: 2